



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Main)

UNIT TEST

25-07-2021

JEE(Main + Advanced) : NURTURE COURSE [PHASE : TEPS, TOPS, I, I(B), II(A) & NURTURE (LIVE) I, I(B), II(A)]

Do not open this Test Booklet until you are asked to do so.

Read carefully the Instructions on this Test Booklet.

Paper : Physics, Chemistry & Mathematics

Important Instructions :

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **90** questions. The maximum marks are **300**.
5. There are **three** parts in the question paper 1,2,3 consisting of **Physics, Chemistry and Mathematics** having **30 questions** in each subject and each subject having **Two sections**.
 - (i) Section-I contains 20 multiple choice questions with **only one correct** option.
Marking scheme : +4 for correct answer, 0 if not attempted and –1 in all other cases.
 - (ii) Section-II contains 10 **Numerical Value Type** questions. Attempt any 5 questions. First 5 attempted questions will be considered for marking.
Marking scheme : +4 for correct answer and 0 in all other cases.
6. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** and **Side-2** of the Answer Sheet. **Use of pencil is strictly prohibited**.
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold or make any stray marks on the Answer Sheet.**

Name of the Candidate (in Capitals) _____

Form Number : in figures _____

: in words _____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ Invigilator's Signature : _____

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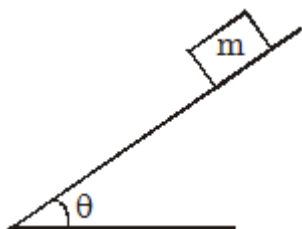
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PART-1 : PHYSICS

SECTION-I : (Maximum Marks: 80)

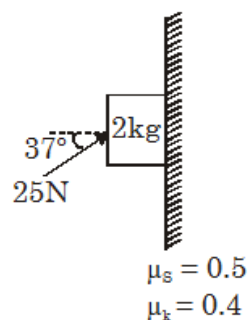
This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:
Full Marks : +4 If correct answer is selected.
Zero Marks : 0 If none of the option is selected.
Negative Marks : -1 If wrong option is selected.

1. In the given figure the coefficient of friction between block and incline is μ . If the block slides on inclined with constant velocity, the friction acting on block will be :-

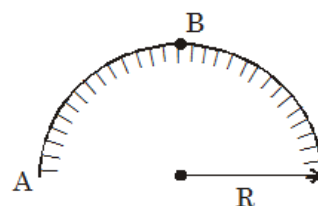


- (A) $\mu mg \sin \theta$
 (B) $\mu mg \tan \theta$
 (C) $mg \sin \theta$
 (D) $> \mu mg \cos \theta$
2. If a particle moves in a circle of radius 4m at speed given by $v = 2t$ where v in m/s & t in sec then at $t = 4$ sec, acceleration of particle will be :
- (A) 16 m/s^2
 (B) $\sqrt{260} \text{ m/s}^2$
 (C) $2\sqrt{17} \text{ m/s}^2$
 (D) 4 m/s^2

3. A 25 N force is applied to the block as shown in figure. Choose the correct option if initial velocity of block is zero (Take $g = 10 \text{ m/s}^2$) :-

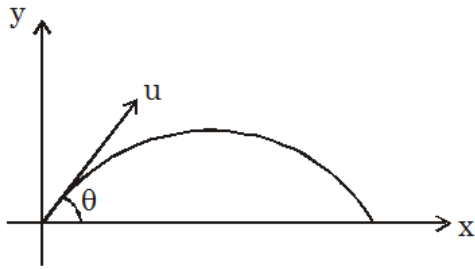


- (A) Normal contact force is 10N.
 (B) Friction force is 8N.
 (C) Friction force is 10N.
 (D) Block is in equilibrium.
4. If a particle starts from A along the curved circular path as shown in figure with constant tangential acceleration a_0 , then acceleration at B in magnitude is :

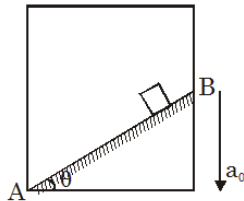


- (A) $2a_0\sqrt{1+\pi^2}$ (B) $a_0\sqrt{\pi^2-1}$
 (C) $a_0\sqrt{1+\pi^2}$ (D) $a_0\pi\sqrt{1+\pi^2}$
5. A particle is moving in a circular path with speed varying with time as $v(\text{m/s}) = 1.5t^2 + 2t$. If the radius of circular path is 2 cm then the angular acceleration at $t = 2\text{sec}$ will be:
- (A) 4 rad/s^2 (B) 400 rad/s^2
 (C) 40 rad/s^2 (D) 0.4 rad/s^2

6. In a projectile motion as shown in figure, during the flight.

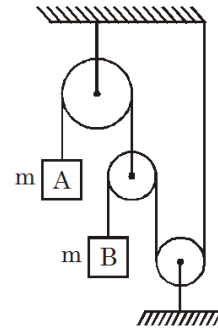


- (A) Magnitude of rate of change of speed at any point is g .
 (B) Magnitude of rate of change of velocity at any point is g .
 (C) Rate of change of speed at top point is g .
 (D) Rate of change of speed at top point is $g/2$.
7. A particle slides down a smooth inclined plane of elevation θ , fixed in a lift going down with constant acceleration ' a_0 '. Length of incline plane AB is ' L ' and particle is released from rest w.r.t. lift from position B at $t = 0$. ($a_0 < g$) :-

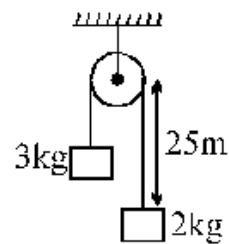


- (A) Acceleration of particle is $g - a_0 \sin \theta$.
 (B) Velocity of the particle when it reaches at point A is $\sqrt{2(g - a_0) \sin \theta L}$
 (C) Time taken by particle to move from B to A is $\sqrt{\frac{2L}{(g - a_0) \sin \theta}}$
 (D) Distance travelled by particle during its motion from B to A is L .

8. In the arrangement as shown in figure, the pulleys are smooth and the strings are inextensible. The acceleration of block B is :

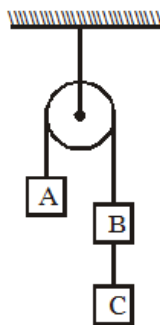


- (A) $\frac{g}{5}$ (B) $\frac{5g}{5}$ (C) $\frac{2g}{5}$ (D) $\frac{2g}{3}$
9. A ball is thrown vertically upwards from the ground. It crosses a point at the height of 10 m twice at an interval of 1 sec. The ball was thrown with the velocity of
 (A) 20m/sec. (B) 25 m/sec
 (C) 30m/sec. (D) 15 m/sec.
10. A particle is moving in a straight line with some initial velocity and constant acceleration. If the sum of the distances travelled in t^{th} & $(t + 1)^{\text{th}}$ sec. is 100 cm, then its velocity after t sec. (in cm/s) is :-
 (A) 100 (B) 25 (C) 75 (D) 50
11. Find the time in which 2 kg mass will strikes the pulley after being released from rest.

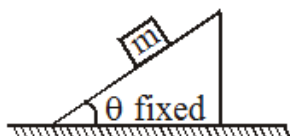


- (A) 10 sec (B) 2 sec (C) 5 sec (D) 4 sec

12. Three blocks A, B and C of mass 10 kg each are hanging on a string passing over a fixed frictionless pulley as shown in figure. The tension in the string connecting blocks B and C is ($g = 10 \text{ m/s}^2$)



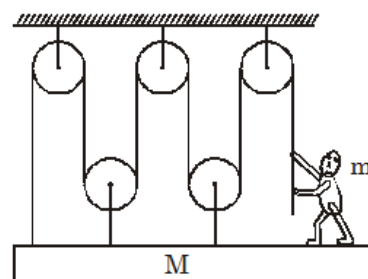
- (A) $\frac{2000}{3} \text{ N}$ (B) $\frac{400}{3} \text{ N}$
(C) 100 N (D) $\frac{200}{3} \text{ N}$
13. Choose the correct equation for normal reaction N between block and fixed smooth wedge, where a is acceleration of block along the incline.



- (A) $N \cos \theta = mg$
(B) $N = mg \cos \theta$
(C) $N \cos \theta - mg = ma \sin \theta$
(D) All are correct.
14. When the kinetic energy of the body is increased by 300 %, the momentum of the body is increased by :-
- (A) 20 % (B) 50 %
(C) 100 % (D) 200 %

15. If $y = \frac{\sqrt{Fx - mv^2}}{\sqrt{m}}$ where $F \rightarrow$ Force, $x \rightarrow$ displacement, $m \rightarrow$ mass, $v \rightarrow$ velocity then y can be :-

- (A) Velocity
(B) Momentum
(C) Angular velocity
(D) Angular momentum
16. In the given figure pulleys and strings are ideal and system (man + plank) is in equilibrium. If man applies a force F on string then choose the necessary condition for system to be in equilibrium. Mass of man is m and mass of plank is M :-

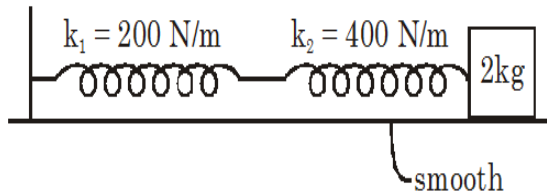


- (A) $F = \frac{(M + m)g}{5}$
(B) $F = \frac{Mg}{5}$
(C) $m \geq \frac{M}{5}$
(D) $m < \frac{M}{5}$

17. A block of mass m is initially at rest on horizontal floor having static friction coefficient μ_s and kinetic friction coefficient is μ_k . If F is the minimum force required to move the block then :-



- (A) Value of F does not depends on μ_s
 (B) Value of F does not depends m
 (C) Value of F does not depends μ_k
 (D) F is slightly less than $\frac{\mu_s mg}{\sqrt{1 + \mu_s^2}}$
18. Block of mass 2kg is displaced by 6 cm towards right and released, springs are initially in natural length. What will be the acceleration of block when it comes to the initial position :-



- (A) 0 (B) 2 m/s^2
 (C) 4 m/s^2 (D) 3 m/s^2
19. If two ends of an ideal rope are pulled with forces of equal magnitude of 100 N in opposite direction, the tension at the center of the rope must be :-
- (A) 200 N (B) 100 N
 (C) 50 N (D) 0 N

20. There are four forces $\vec{F}_1, \vec{F}_2, \vec{F}_3, \vec{F}_4$ acting on particle such that particle is in equilibrium. Suddenly $\vec{F}_4$ vanishes. The resultant of remaining forces acting on particle is :-

- (A) $\vec{F}_1 + \vec{F}_2$
 (B) $F_1 + F_2 + F_3$
 (C) $\vec{F}_1 + \vec{F}_2 + \vec{F}_3$
 (D) $\vec{F}_4$

SECTION-II : (Maximum Marks: 40)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

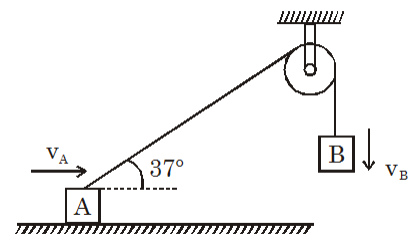
For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the second decimal place; e.g. 6.25, 7.00, -0.33 , $-.30$, 30.27, -127.30 , if answer is 11.36777..... then both 11.36 and 11.37 will be correct)

Answer to each question will be evaluated according to the following marking scheme:

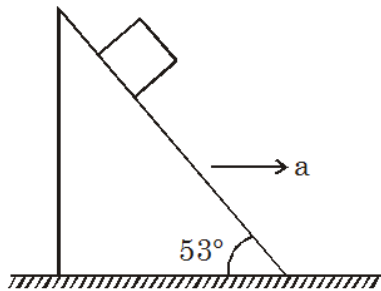
Full Marks : +4 If ONLY the correct numerical value is entered as answer.

Zero Marks : 0 In all other cases.

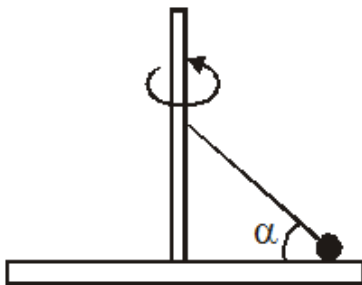
1. If velocity of block A is 5 m/s . Find velocity of block B in m/s .



2. If acceleration (in m/s^2) of wedge (rightwards) so that block does not move w.r.t. wedge is $\frac{8n}{3}$. Find value of n . (Take $g = 10 \text{ m/s}^2$)

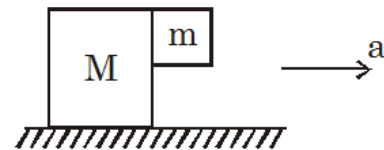


3. A circular platform rotates around a vertical axis with angular velocity $\omega = 10 \text{ rad/s}$. On the platform is a ball of mass 2 kg , attached to the long axis of the platform by a thin rod of length 16 cm ($\alpha = 30^\circ$). Find normal force exerted by the ball on the platform (in newton). Friction is absent.

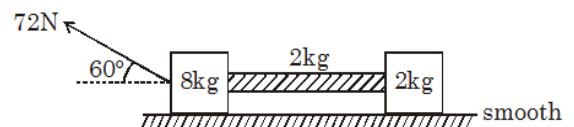


4. A rod is rotating with an angular velocity of $3\pi \text{ rad/s}$. The angle rotated by the rod from $t = 0$ to the time when it has an angular velocity of $24\pi \text{ rad/s}$ is given by $\alpha\pi$ radian. Fill $\frac{\alpha}{2}$ in OMR sheet.

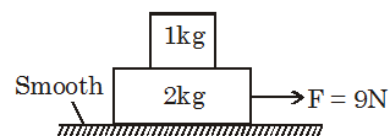
5. The minimum stopping distance on level ground is 5 m for a vehicle moving at 10 m/s . If the same vehicle is going down on incline of inclination 37° . What is the minimum stopping distance (in m) ?
6. With what minimum acceleration (in m/s^2) mass M must be moved on frictionless surface so that m remains at rest with respect to 'M' as shown. The coefficient of friction between M and m is 0.4 . ($g = 10 \text{ m/s}^2$)



7. In the figure shown if mass of the rope is 2 kg , then tension (in Newton) at the midpoint of the rope is



8. A car moving with a speed of 20 km/hr can be stopped by applying brakes after at least 2 m . If the same car is moving with a speed of 40 km/hr , what is the minimum stopping distance
9. Compute the least acceleration (in m/s^2) with which a 50 kg man can slide down a rope, if the rope can withstand a tension of 300 N .
10. In the figure, there is no relative motion between the two blocks. Force of friction acting on 1 kg block (in N) is :-



SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

- Select the correct order of electron affinity ?
(A) $F^+ > Cl^+ > Cl > F$ (B) $Cl > Cl^+ > F > F^+$
(C) $Cl > F > F^+ > Cl^+$ (D) $Cl > F > Cl^+ > F^+$
- In which of the following process energy is liberated :-
(A) $Cl \rightarrow Cl^+ + e^\ominus$
(B) $NaCl \rightarrow Na^\oplus + Cl^\ominus$
(C) $Cl + e^\ominus \rightarrow Cl^\ominus$
(D) $O^- + e^\ominus \rightarrow O^{2-}$
- Match the list:

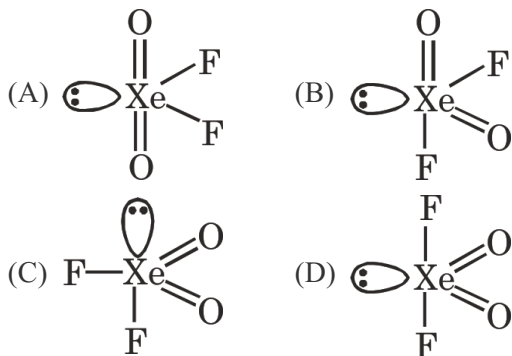
List- I (Order)		List- II (Properties)	
(P)	$Fe^{3+} > Fe^{2+}$	(1)	Radii
(Q)	$O > S$	(2)	Ionisation energy
(R)	$P > N$	(3)	Electron affinity
(S)	$I > I^+$	(4)	Electronegativity

The correct option is

- (A) $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 3; S \rightarrow 1$
 (B) $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 1; S \rightarrow 2$
 (C) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 3$
 (D) $P \rightarrow 1; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 4$

- If m = magnetic quantum number and ℓ = azimuthal quantum number, then relation between given value of ℓ & total possible values of m is -
(A) $m = \ell + 2$
(B) $m = 2\ell^2 + 1$
(C) $\ell = \frac{m-1}{2}$
(D) $\ell = 2m + 1$
- Which of the following statement is **INCORRECT** ?
(A) Total number of maximum of electrons which can have $n = 5, s = +1/2$ and $n = 5, s = -1/2$ are same.
(B) In H-like atoms, energy of the subshells are different in same shell.
(C) The energy order of subshells in only multielectronic atoms is decided by $(n + \ell)$ rule.
(D) None of these
- Select the **CORRECT** statement
(A) A molecule which have T.B.P. geometry axial bond length always greater than that of equatorial bond length
(B) In PCl_3F_2 molecule equatorial bond length is greater than that of axial bond length
(C) A molecule which have P.B.P. geometry, axial orbital length of central atom is greater than that of equatorial orbital length
(D) A molecule which have T.B.P. geometry, equatorial orbital length of central atom is greater than that of axial orbital length

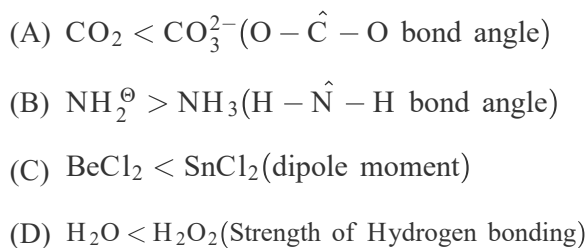
7. The most stable structural representation of XeO_2F_2 molecule according to VSEPR theory is :-



8. In which of the following molecule, number of $p\pi-d\pi$ bonds formed are two times the number of $p\pi-p\pi$ bonds?



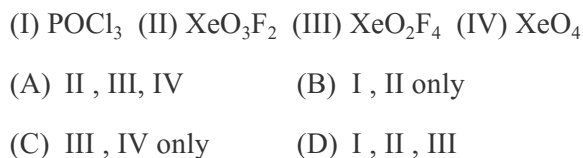
9. Which of the following order is correct.



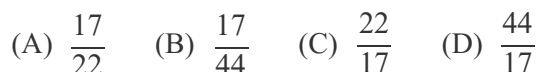
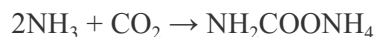
10. In which of the following compound d-orbital is not involved in hybridisation :-



11. In which of the following species presence of double bond does not affect idealised bond angle :-



12. Mass ratio of NH_3 and CO_2 for maximum product formation as per reaction :



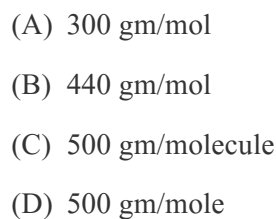
13. Number of neutrons in 5.5 gm T_2O (T is ^3H) are -



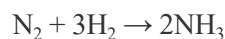
14. What volume of a liquid substance A contains $2N_A$ molecules of A if its molecular weight is 240 and its density is 1.2 g/ml :



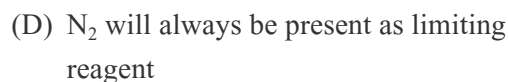
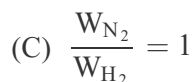
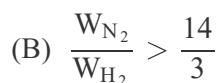
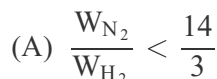
15. A compound contain 3.2% of oxygen. The minimum molar mass of the compound is -



16. Consider the following reaction -



which condition will make H_2 a limiting reagent under all cases (where 'w' is weight of substance)



17. Find $[VD]_{\text{mix}}$ of a gaseous mixture of O_3 and O_2 . Molar ratio of O_3 to O_2 is 1 : 2
(A) $\frac{56}{6}$ (B) $\frac{112}{3}$ (C) $\frac{224}{3}$ (D) $\frac{56}{3}$
18. 90 gm mixture of H_2 and O_2 is taken in stoichiometric ratio and gives H_2O with 80% yield. The produced mass of H_2O (in gm) is :
(A) 45 gm (B) 36 gm (C) 72 gm (D) 90 gm
19. Consider the following reaction
 $NH_3 + O_2 \rightarrow NO_2 + H_2O$
Find sum of mass of all the product (in gm), if 68 g NH_3 reacts with 256 g O_2 in a closed container. [N = 14, O = 16]
(A) 184 (B) 108 (C) 292 (D) 68
20. 3×10^{20} molecules of SO_3 are added to 160 mg of SO_3 , then number of moles of SO_3 finally present: ($N_A = 6 \times 10^{23}$) (S = 32)
(A) 1.5×10^{-3} (B) 3×10^{-3}
(C) 2.5×10^{-3} (D) 1.5×10^{-4}

SECTION-II : (Maximum Marks: 40)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the second decimal place; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered as answer.

Zero Marks : 0 In all other cases.

- Count the number of electrons of 'Zn' which are present in orbitals which has only one nodal plane.
- Find total number of maximum orbitals in 'Sc' atom, for which $|m| \leq 1$ and contains at least one electron.
[At. No. of Sc (z = 21)]
- Total number of elements which can accommodate in the 10th period of periodic table if present.
- How many molecules from the following have different bond angle from ideal bond angle according to hybridisation.
 PCl_3F_2 , CH_2F_2 , H_2O , XeF_4 , IF_5 , SF_4
- What is the % ionic character in HI ? Given electronegativity of H = 2.1 and I = 2.5.
(Give your answer to the nearest integer)
- $BF_3 + NH_3 \rightarrow \text{Product A}$
Total number of lone pair present in product (A)
- Calculate the percent loss in weight after complete decomposition of a pure sample of potassium chlorate. (K = 39, Cl = 35.5)
 $KClO_3(s) \rightarrow KCl(s) + O_2(g)$
- 12 gm of an alkaline earth metal reacts with 8 gm of oxygen gas to form its oxide. Find the atomic weight of metal -
- Find the percentage of carbon by mole of atoms in CH_4 .
- A sample of oxygen gas (only O_2 molecules) contains O^{16} and O^{17} oxygen atoms only in 9 : 1 atomic ratio. The average molecular mass of O_2 in the sample is.

SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

- Sum of all the real solution(s) of equation $9^{\log_3(\log_2 x)} = \log_2 x - \log_2^2 x + 1$ is -
(A) a prime number
(B) an irrational number
(C) a rational number which is not integer
(D) a composite number
- If $(xyz)^{(x+y+z)} = 1$ where $xyz \neq 1$ and $\log_3(xyz) = -1$, then $\log_{(xyz)}(x^3 + y^3 + z^3)$ equals to :
(A) 0 (B) 1
(C) -1 (D) 2
- If $a_0 = \cos\theta$ where $\theta \in \left(0, \frac{\pi}{2}\right)$ and $a_{r+1} = \sqrt{\frac{1}{2}(1 + a_r)}$; $\forall r \in \{0, 1, 2, 3, \dots\}$, then $\frac{\sqrt{1 - a_0^2}}{(a_1 \cdot a_2 \cdot a_3 \cdot \dots \cdot a_n) \sqrt{1 - a_n^2}}$ is equal to
(A) 2^n
(B) $\frac{1}{2^n}$
(C) $2^n \sin \frac{\theta}{2^n}$
(D) $\frac{1}{2^n} \sin \frac{\theta}{2^n}$

- If the range of the function $f(x) = \frac{1}{3 \sin x - 4 \cos x + 2}$ is $(-\infty, b] \cup [c, \infty)$ then the value of $\frac{1}{b} + \frac{1}{c}$ is equal to
(A) 10 (B) 4
(C) $-\frac{4}{21}$ (D) $\frac{4}{21}$
- Let $ax^2 + (b + 2)x + c = 0$ and $x^2 + 5x + 2 = 0$ has a common root, where $a, b, c \in \mathbb{N}$, then minimum value of $(a + b + c)$ is
(A) 3 (B) 4 (C) 5 (D) 6
- Number of integral roots of the equation $x^8 - 2x^7 - 8x^5 + 7x^2 + 77 = 0$ is equal to
(A) 3 (B) 2 (C) 1 (D) 0
- The number of solution(s) of the equation $3\cos^2 x = 8\sin x$ in $[0, \pi]$ is-
(A) 1 (B) 2 (C) 3 (D) 4
- If $\lambda \tan^2 x + \frac{\cot^2 x}{\lambda} = 2$ where $\lambda > 0$ & $x \in \left(0, \frac{\pi}{2}\right)$, then $\sin^2 x + 5\cos^2 x + 3\sin x \cos x$ is equal to -
(A) $\frac{1}{(\lambda + 1)}$ (B) $\frac{(1 - 5\lambda)}{(\lambda + 1)}$
(C) $\frac{1 - 5\lambda + 3\sqrt{\lambda}}{(\lambda + 1)}$ (D) $\frac{1 + 5\lambda + 3\sqrt{\lambda}}{(\lambda + 1)}$
- If $a_1, a_2, \dots, a_{15}$ are in arithmetic progression and $a_1 + a_8 + a_{15} = 15$, then $a_2 + a_3 + a_8 + a_{13} + a_{14}$ is equal to :-
(A) 25 (B) 35
(C) 10 (D) 15

10. Number of positive integral solutions of the equation $\frac{x}{3} + \frac{14}{y} = 3$, is
(A) 2 (B) 3 (C) 7 (D) 5
11. If $a > b > c$ and $a^3 + b^3 + c^3 = 3abc$, then for the equation $ax^2 + bx + c = 0$, $a \neq 0$ which of the following is true ?
(A) Both roots are real and equal
(B) One root must be positive and other must be negative
(C) Both roots must be real and positive
(D) Both roots must be negative and real
12. Minimum value of $9\sec^2\theta + 4\operatorname{cosec}^2\theta$ is
(A) 12 (B) 25 (C) 13 (D) 20
13. If $\sin x + \sin^2 x = 1$ then the value of $\cos^2 x + \cos^4 x$ is-
(A) 0 (B) 2
(C) 1 (D) None of these
14. If the maximum value of $y = \frac{7 + 6 \tan x - \tan^2 x}{(1 + \tan^2 x)}$ is λ , then the value of $\log_{\sqrt{2}}(\lambda)$ is :-
(A) 0 (B) 6 (C) 8 (D) 1
15. $\sin 47^\circ + \sin 61^\circ - \sin 11^\circ - \sin 25^\circ$ is equal to
(A) $\sin 36^\circ$
(B) $\cos 36^\circ$
(C) $\sin 7^\circ$
(D) $\cos 7^\circ$
16. Number of solutions of the equation $\left(\frac{1}{\sqrt{2}}\right)^{\tan x} = (2)^{\sin x}$ in the interval $x \in [0, \pi] - \left\{\frac{\pi}{2}\right\}$ is
(A) 1
(B) 2
(C) 3
(D) More than 3
17. The value of $\sec 80^\circ - \sqrt{3} \operatorname{cosec} 80^\circ$ is
(A) 4 (B) 1 (C) 2 (D) 16
18. If the ratio of sum of n terms of two A.P.'s be $5n + 3 : 7n + 1$, then the ratio of their 7th term is-
(A) 53 : 12
(B) 17 : 23
(C) 19 : 18
(D) 13 : 19
19. Consider a sequence $\{a_n\}$ with $a_1 = 2$ & $a_n = \frac{a_{n-1}^2}{a_{n-2}}$ $\forall n \geq 3$. If a_2 & a_5 are +ve integers & $a_5 \leq 512$, then number of possible value of a_5 are -
(A) 1 (B) 2 (C) 3 (D) 4
20. $\tan 3^\circ + 2 \tan 6^\circ + 4 \tan 12^\circ + 8 \cot 24^\circ = \cot \theta^\circ$ ($0 < \theta < 90$), then
(A) $\cot (10 \theta)^\circ = 1$
(B) $\cot (15 \theta)^\circ = 1$
(C) $\cot \theta^\circ = 0$
(D) $\cot (15 \theta)^\circ = \sqrt{3}$

SECTION-II : (Maximum Marks: 40)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the second decimal place; e.g. 6.25, 7.00, -0.33, -30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered as answer.

Zero Marks : 0 In all other cases.

1. If α & β are roots of the equation $x^2 + x - 3 = 0$, then value of $\alpha^3 - 4\beta^2 + 24$ is
2. Let $\{t_n\}$ be a sequence of integers in G.P. in which $t_3 : t_6 = 1 : 8$ and $t_1 + t_5 = 170$ then t_2 is-
3. If $\sin\alpha + \sin\beta = 2$, then $6\cos(\alpha - \beta)$ equals to

4. The value of $\sin^2 x + \sin^2\left(x + \frac{\pi}{3}\right) + \cos x \cos\left(x + \frac{\pi}{3}\right)$ is :-
5. If $\sin\theta, \sin^2\theta, 1$ are the first three terms of non constant AP, then number of value(s) of θ in $(-\pi, \pi)$ is
6. Let $a_1, a_2, a_3, \dots, a_{10}$ be in A.P. and $h_1, h_2, h_3, \dots, h_{10}$ be in H.P. If $a_1 = h_1 = 2$ and $a_{10} = h_{10} = 3$ then the value of $\frac{a_3 h_8}{4}$ is
7. Maximum possible value of the expression $y = \sqrt{\sin^4 x + 4\cos^2 x} - \sqrt{\cos^4 x + 4\sin^2 x}$ is :-
8. If $\sin 2\alpha = 4 \sin 2\beta$ and $5 \tan(\alpha - \beta) = k \tan(\alpha + \beta)$, then the value of k is ?
9. Number of integral values of 'a' such that $x^2 + ax + a + 1 = 0$ has integral roots is equal to
10. If a, b, c are positive real numbers then the number of real roots of the equation $ax^2 + b|x| + c = 0$ is

Space for Rough Work